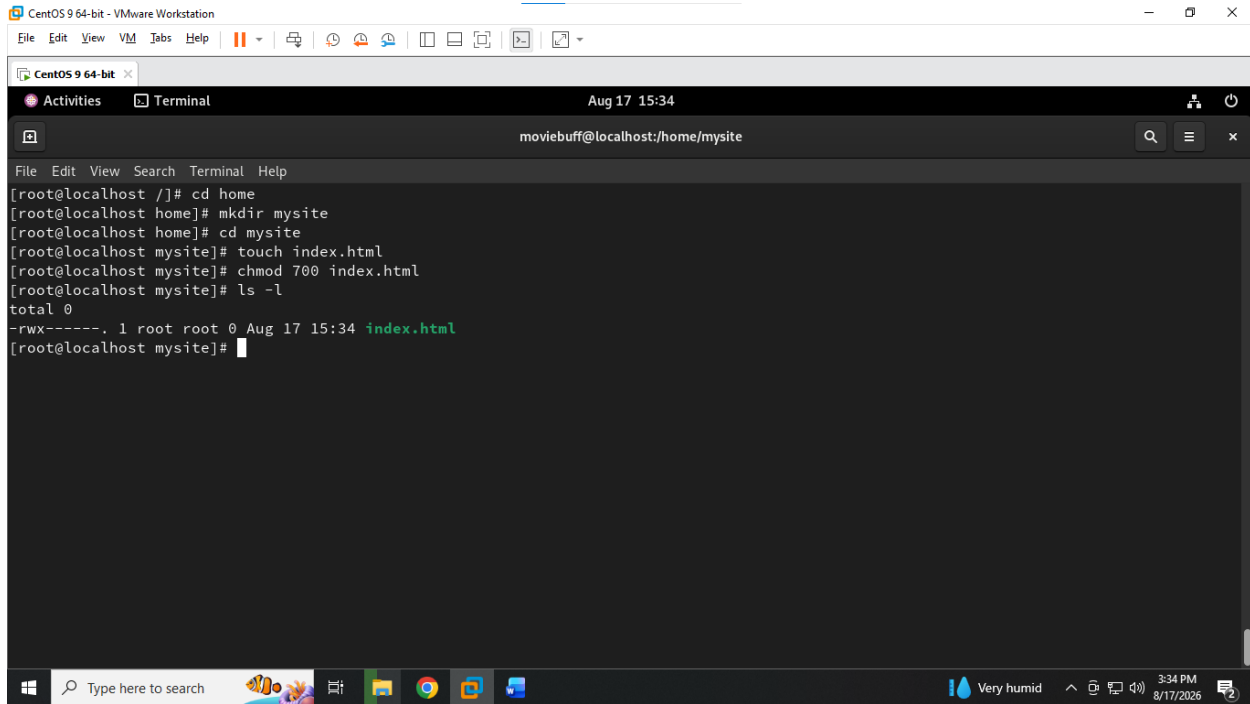


Module-4 : Introduction to Linux Services

Guided by : Vipul mistri

Q.1 Create a new directory called mysite in your home folder, place an index.html file inside it, and set permissions so only your user can read and access the site files.

ANS :



The screenshot shows a terminal window titled 'CentOS 9 64-bit' within a VMware Workstation environment. The terminal output shows the following commands and results:

```
[root@localhost ~]# cd home
[root@localhost home]# mkdir mysite
[root@localhost home]# cd mysite
[root@localhost mysite]# touch index.html
[root@localhost mysite]# chmod 700 index.html
[root@localhost mysite]# ls -l
total 0
-rwx-----. 1 root root 0 Aug 17 15:34 index.html
[root@localhost mysite]#
```

The terminal window has a menu bar with 'File', 'Edit', 'View', 'Search', 'Terminal', and 'Help'. The status bar at the bottom shows the system time as 3:34 PM on 8/17/2026, along with weather information 'Very humid' and a notification icon.

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ANS :

The screenshot shows a Windows 10 desktop with a VMware Workstation window titled 'CentOS 9 64-bit - VMware Workstation'. The virtual machine is running, and the terminal window is open, showing the following commands and output:

```

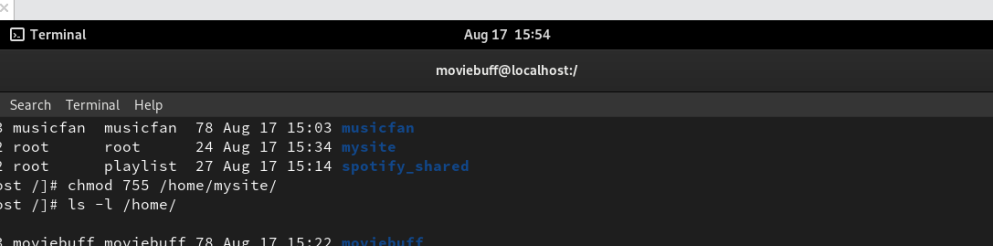
[root@localhost httpd]# cd /
[root@localhost /]# ls
afs bin boot dev etc home lib lib64 media mnt opt proc root run sbin srv sys tmp usr var
[root@localhost /]# ls home/
moviebuff/ musicfan/ mysite/ spotify_shared/
[root@localhost /]# ls home/mysite/
index.html
[root@localhost /]# ls -l /home/mysite/
total 0
-rw-r--r--. 1 root root 0 Aug 17 15:34 index.html
[root@localhost /]# ls -l /home/
total 0
drwx-----. 3 moviebuff moviebuff 78 Aug 17 15:22 moviebuff
drwx-----. 3 musicfan musicfan 78 Aug 17 15:03 musicfan
drwxr-xr-x. 2 root root 24 Aug 17 15:34 mysite
drwxrw----. 2 root playlist 27 Aug 17 15:14 spotify_shared
[root@localhost /]# chmod 755 /home/mysite/
[root@localhost /]# ls -l /home/
total 0
drwx-----. 3 moviebuff moviebuff 78 Aug 17 15:22 moviebuff
drwx-----. 3 musicfan musicfan 78 Aug 17 15:03 musicfan
drwxr-xr-x. 2 root root 24 Aug 17 15:34 mysite
drwxrw----. 2 root playlist 27 Aug 17 15:14 spotify_shared
[root@localhost /]#

```

The terminal window is titled 'CentOS 9 64-bit x' and shows the user 'moviebuff@localhost:/'.

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```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Aug 17 15:54
moviebuff@localhost:/
File Edit View Search Terminal Help
drwx----- 3 musicfan musicfan 78 Aug 17 15:03 musicfan
drwxr-xr-x 2 root root 24 Aug 17 15:34 mysite
drwxrw---- 2 root playlist 27 Aug 17 15:14 spotify_shared
[root@localhost ~]# chmod 755 /home/mysite/
[root@localhost ~]# ls -l /home/
total 0
drwx----- 3 moviebuff moviebuff 78 Aug 17 15:22 moviebuff
drwx----- 3 musicfan musicfan 78 Aug 17 15:03 musicfan
drwxr-xr-x 2 root root 24 Aug 17 15:34 mysite
drwxrw---- 2 root playlist 27 Aug 17 15:14 spotify_shared
[root@localhost ~]# vim /home/mysite/index.html
[root@localhost ~]# vim /home/mysite/index.html
[root@localhost ~]# curl http://mysite.local
<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>403 Forbidden</title>
</head><body>
<h1>Forbidden</h1>
<p>You don't have permission to access this resource.</p>
</body></html>
[root@localhost ~]# getenforce
Enforcing
[root@localhost ~]# chcon -R -t httpd_sys_content_t /home/mysite
[root@localhost ~]# systemctl restart httpd
[root@localhost ~]#
```

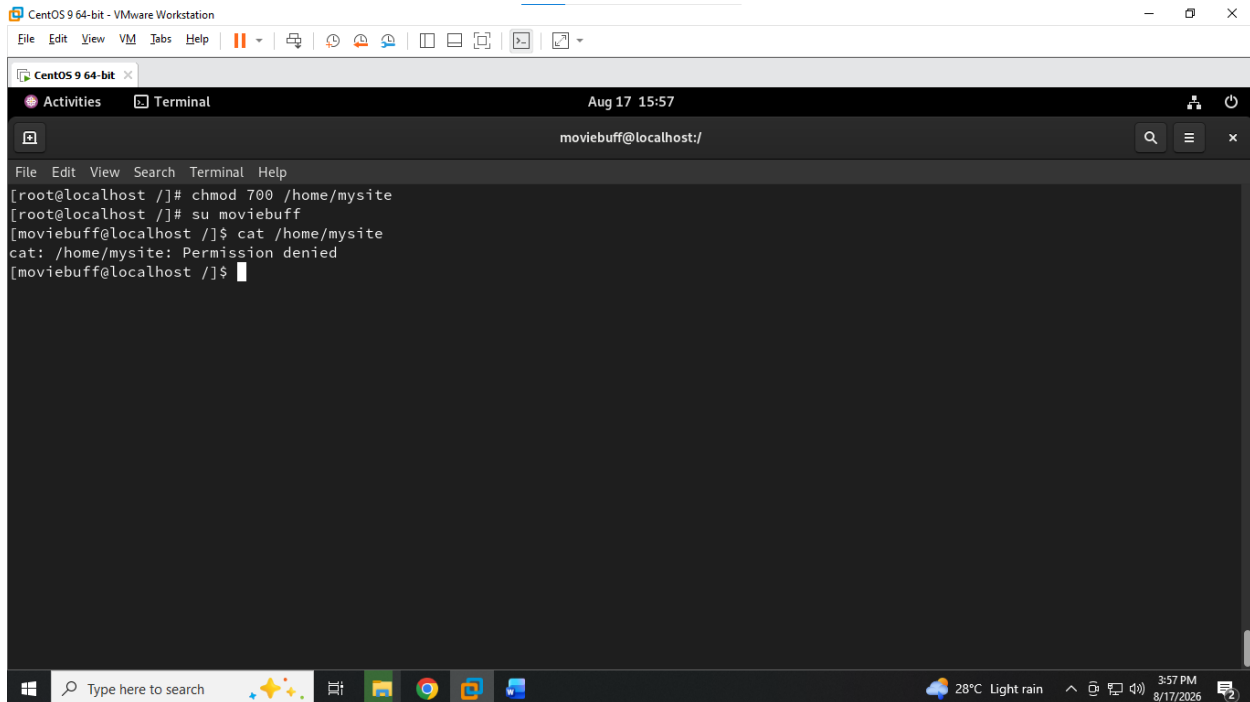
The screenshot shows a VMware Workstation interface with a single virtual machine named 'CentOS 9 64-bit' running. The virtual machine's desktop environment is visible, featuring a black background with a white terminal window. The terminal displays the text 'Welcome to My Site' in a large, bold, serif font, followed by 'Apache is serving mysite successfully.' in a smaller, regular font. The top of the window shows the VMware menu bar and a toolbar. The bottom of the window shows the Windows taskbar with various icons and the system clock indicating 3:55 PM on 8/17/2026.

Module-4 : Introduction to Linux Services

Guided by : Vipul mistri

Q.3 Change the permissions of the mysite directory so that another user on the system cannot access the site files directly. Try to access the files as the other user and note the result. Hint: Use chmod and su or a different terminal window for the second user.

ANS :



The screenshot shows a terminal window titled 'CentOS 9 64-bit' running within a VMware Workstation. The terminal output is as follows:

```
[root@localhost ~]# chmod 700 /home/mysite
[root@localhost ~]# su moviebuff
[moviebuff@localhost ~]$ cat /home/mysite
cat: /home/mysite: Permission denied
[moviebuff@localhost ~]$
```

The terminal window has a menu bar with 'File', 'Edit', 'View', 'Search', 'Terminal', and 'Help'. The status bar at the bottom shows the date and time as 'Aug 17 15:57' and the user as 'moviebuff@localhost:/'.

Permission denied because we have given 700 means only owner have full permission other and groups have no permission to access read write or execute this folder.

Q.4 Document the steps you took to restrict access to your hosted site, including the commands used, permission settings, and screenshots or outputs showing successful and failed access attempts.

ANS : **Restricting Access to the Hosted Apache Website**

Objective

The objective was to configure an Apache-hosted website and control access to the website using Linux file and directory permissions. The website was hosted from /home/mysite.

1. Website Directory

The website was created in:

/home/mysite

The directory contained:

index.html

The Apache configuration was created at:

/etc/httpd/conf.d/mysite.conf

The configuration used:

```
<VirtualHost *:80>
```

```
    ServerName mysite.local
```

```
    DocumentRoot /home/mysite
```

```
    <Directory /home/mysite>
```

```
        Require all granted
```

```
    </Directory>
```

```
</VirtualHost>
```

2. Checking Apache Configuration

The configuration was tested with:

```
sudo apachectl configtest
```

The output was:

Syntax OK

Apache was then restarted:

```
systemctl restart httpd
```

3. Configuring Hostname Access

The hostname was added to /etc/hosts:

```
127.0.0.1    mysite.local
```

This allowed the site to be accessed through:

```
http://mysite.local
```

4. Permission Settings

The website directory initially required permission changes so Apache could access the files.

The directory permission was set to:

```
chmod 755 /home/mysite
```

The HTML file permission was set to:

```
chmod 644 /home/mysite/index.html
```

The 755 permission allows the directory to be traversed and read, while 644 allows the HTML file to be read by Apache without giving other users write permission.

5. Failed Access Attempt

After configuring the hostname, Firefox successfully reached mysite.local, but the browser displayed:

```
403 Forbidden
```

You don't have permission to access this resource.

This demonstrated that Apache was running and the hostname was resolving, but access to /home/mysite was still being denied because of the filesystem permissions.

The directory permissions were then corrected using:

```
chmod 755 /home/mysite
```

and the HTML file was configured with:

```
chmod 644 /home/mysite/index.html
```

6. SELinux Access

Because CentOS uses SELinux, its status was checked with:

```
getenforce
```

If SELinux was enforcing, the appropriate Apache content context was applied:

```
chcon -R -t httpd_sys_content_t /home/mysite
```

Apache was then restarted:

```
systemctl restart httpd
```

7. Access Verification

The website could be tested from the terminal using:

```
curl -I http://mysite.local
```

A successful configuration should return:

```
HTTP/1.1 200 OK
```

The website can also be verified in Firefox by opening:

```
http://mysite.local
```

Conclusion

The Apache website was configured to serve /home/mysite using a VirtualHost. Access was controlled through Linux directory and file permissions. The failed browser attempt produced a **403 Forbidden** error, demonstrating that Apache initially could not access the website content. After correcting the permissions and, where necessary, the SELinux context, Apache was able to access the hosted content. The Apache configuration itself was verified successfully with the **Syntax OK** output.